

# ***Banks and Agencies***

## ***Database Project***

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### **1. Domain Description**

An banking agency, or agency bank, is a branch that does business on behalf of some other bank. These agencies may be used to help corporations expand overseas, arrange loans, or manage funds.

This project simulates the database system of the Central Tunisian Bank handling multiple banks and branches (agencies). It handles customer management, employee tracking, accounts, transactions, loans, credit cards, and support services. This system ensures data integrity, efficient query support, and scalability.

### **2. Environment Analysis**

A banking system operates within a complex environment that includes technological, regulatory, customer-focused, and financial elements.

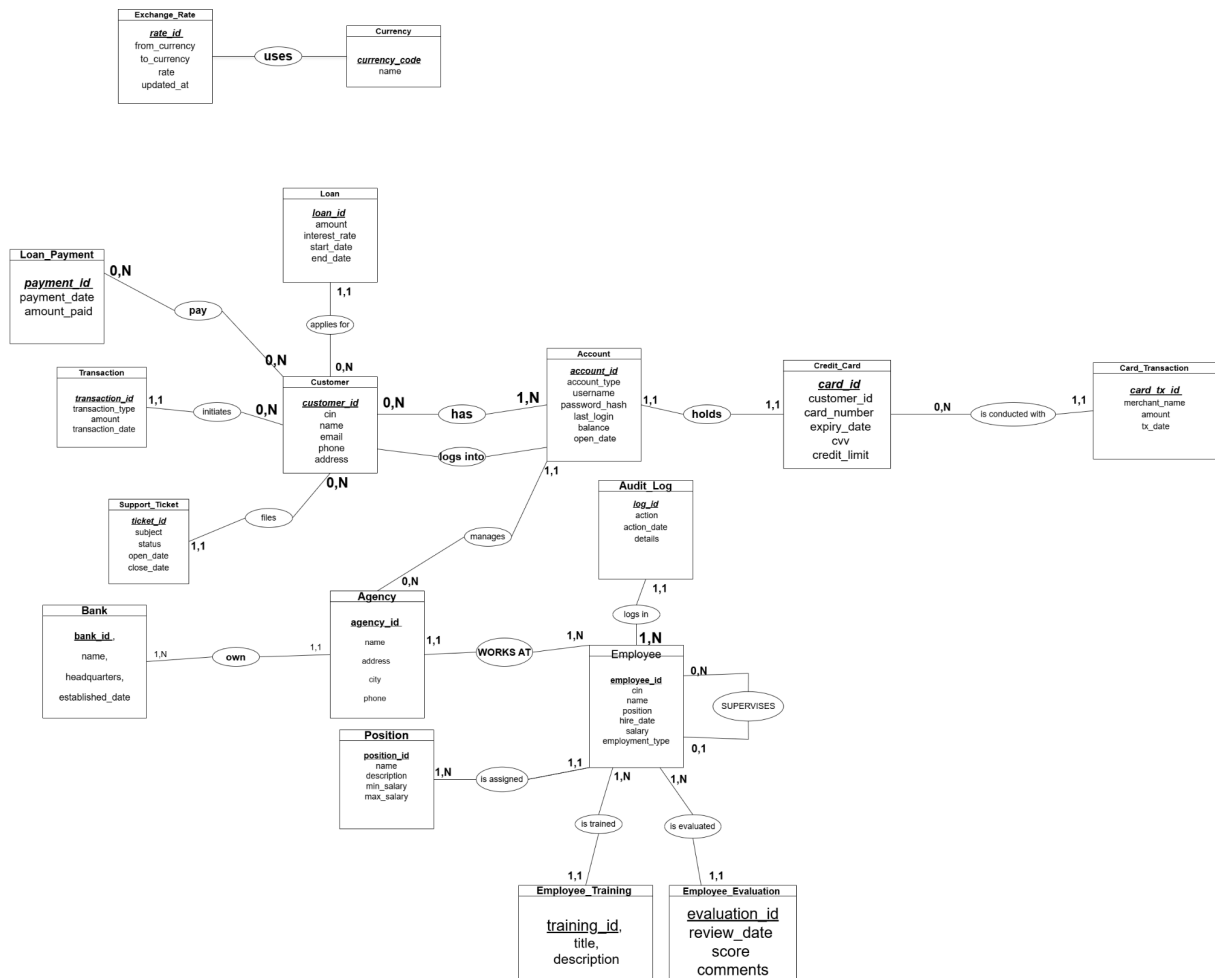
- Banks operate under **strict government regulation** (e.g. Central Bank policies, anti-money laundering laws).
- Licensing and auditing are government-controlled.
- Interest rates, inflation, and exchange rates directly impact banking operations.
- Customers require **loan services, savings accounts, and credit management..**
- Customers demand **easy access** to accounts via online/mobile banking.
- Systems must support **secure transactions, multi-channel access** (web/mobile), and **automated audits**.
- Integration with national and international **currency exchange** systems is crucial.

- Increasing pressure on banks to **reduce paper use** and **optimize digital services**.
- Some banks promote **green investments** or eco-loans.
- Systems must maintain **logs of employee actions, support tickets, and transaction records**.

After analysing the environment, we decided to choose these entities to work on:

- Bank
- Agency
- Customer
- Employee
- Account
- Transaction
- Loan
- Loan Payment
- Credit Card
- Card Transaction
- User Login
- Support Ticket
- Currency
- Exchange Rate
- Audit Log

### 3. Conceptual Design (ER Diagram Description)



\*This design will be attached with the files

## • Entities and Their Attributes

| Entity Name         | Attributes                                                                        |
|---------------------|-----------------------------------------------------------------------------------|
| Bank                | bank_id, name, headquarters, established_date                                     |
| Agency              | agency_id, name, address, city                                                    |
| Employee            | employee_id, name, position, hire_date, salary, employment_type                   |
| Position            | position_id, name, description, min_salary, max_salary                            |
| Customer            | customer_id, cin, name, email, phone, address                                     |
| Account             | account_id, account_type, username, password_hash, last_login, balance, open_date |
| Transaction         | transaction_id, transaction_type, amount, transaction_date                        |
| Loan                | loan_id, amount, interest_rate, start_date, end_date                              |
| Loan_Payment        | payment_id, payment_date, amount_paid                                             |
| Credit_Card         | card_id, customer_id, card_number, expiry_date, CVV, credit_limit                 |
| Card_Transaction    | card_tx_id, merchant_name, amount, tx_date                                        |
| Currency            | currency_code, name                                                               |
| Exchange_Rate       | rate_id, from_currency, to_currency, rate, updated_at                             |
| Support_Ticket      | ticket_id, subject, status, open_date, close_date                                 |
| Audit_Log           | log_id, action, action_date, details                                              |
| Employee_Training   | training_id, title, description                                                   |
| Employee_Evaluation | evaluation_id, review_date, score, comments                                       |

## . Entity Relationships and Cardinalities

| Relationship      | Entities Involved              | Cardinality |
|-------------------|--------------------------------|-------------|
| own               | Bank → Agency                  | 1,1 - 1,N   |
| works at          | Employee → Agency              | 1,N - 1,1   |
| is assigned       | Employee → Position            | 1,N - 1,1   |
| is trained        | Employee → Employee_Training   | 1,N - 1,1   |
| is evaluated      | Employee → Employee_Evaluation | 0,1 - 1,1   |
| supervises        | Employee → Employee            | 0,N - 0,1   |
| logs in           | Employee → Audit_Log           | 1,1 - 1,N   |
| manages           | Agency → Account               | 1,1 - 1,N   |
| has               | Customer → Account             | 0,N - 1,1   |
| logs into         | Customer → Account             | 0,N - 1,1   |
| initiates         | Customer → Transaction         | 0,N - 1,1   |
| applies for       | Customer → Loan                | 0,N - 1,1   |
| pays              | Loan_Payment → Loan            | 0,N - 1,1   |
| holds             | Customer → Credit_Card         | 1,N - 1,1   |
| is conducted with | Credit_Card → Card_Transaction | 0,N - 1,1   |
| files             | Customer → Support_Ticket      | 0,N - 1,1   |
| uses              | Exchange_Rate → Currency       | 1,1 - 1,1   |

## 4. Logical Design (Relational Model) :

**Bank**( bank\_id , name, headquarters, established\_date )

**Agency**( agency\_id , #bank\_id, name, address, city, phone )

**Employee**( employee\_id , #agency\_id, #position\_id, name,cin,  
hire\_date, salary, employment\_type )

**Position**( position\_id , name, description, min\_salary, max\_salary )

**Employee\_Training**( training\_id , title, description )

**Employee\_Training\_Assignment**(training\_assignment\_id, #employee\_id,  
#training\_id)

**Employee\_Evaluation**( evaluation\_id , #employee\_id, review\_date,  
score, comments )

**Employee\_Supervision**(supervision\_id, #supervisor\_id,  
#subordinate\_id)

**Customer**( customer\_id ,cin, name, email, phone, address)

**Account**( account\_id , #customer\_id, account\_type, username,  
password\_hash, last\_login, balance, open\_date )

**Transaction**(transaction\_id , #account\_id, transaction\_type, amount,  
transaction\_date )

**Loan**( loan\_id , #customer\_id, amount, interest\_rate, start\_date,  
end\_date )

**Loan\_Payment**( payment\_id , #loan\_id, payment\_date, amount\_paid )

**Credit\_Card**( card\_id , #customer\_id, card\_number, expiry\_date, cvv,  
credit\_limit )

**Card\_Transaction**( card\_tx\_id , #card\_id, merchant\_name, amount,  
tx\_date )

**Currency**( currency\_code , name )

**Exchange\_Rate**( rate\_id , #from\_currency, #to\_currency, rate,  
updated\_at )

**Support\_Ticket**( ticket\_id , #customer\_id, subject, status, open\_date, close\_date )

**Audit\_Log**( log\_id , #employee\_id, action, action\_date, details )

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## 5. Normalization

All tables are normalized to 3NF:

- 1NF: Each column contains atomic values.
- 2NF: All non-key attributes are fully functionally dependent on the whole primary key.
- 3NF: No transitive dependencies exist.

## 6. Implementation

The tables mentioned in the logical design were implemented in SQL following constraints , the constraints used were :

1. **Primary Key**
2. **Foreign Key**
3. **Not Null**
4. **Unique**
5. **Check Constraint**
  - Salaries and balances can't be negative.
  - Interest rates and loan amounts must be positive.
  - CVV codes must be exactly 3 digits.
  - start\_date can't be after the end\_date
  - Employees can't supervise themselves

## 7. Insertion

The inserted values follows 3 imaginary customers (Ali,Salah,Amine) and 3 imaginairy employees (Rayen, Ahmed, Ilef) and the other values are random

A .sql file contains the code for this implementation